FAR.AI · PRE-TRAINING SAFETY RESEARCH LEAD · TEST TASK

Distillation-time interventions

A 12-month pre-training safety agenda, and a toy experiment that derisks its load-bearing assumption

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BEFORE THE AGENDA

Two framing notes

1

The boundaries are fuzzy

Data filtering, gradient routing, unlearning and synthetic data blur into one another. The genres within pre-training safety are useful labels, not walls.

2

Allocate effort as a portfolio

Some bets should be speculative and greenfield; others should be practical, like scaling what already works. A team needs both.

Distillation, defined broadly

> *“Training model A on the outputs of model B.”*

Why people do it

It recreates a capability of B inside A for less compute than building that capability in A from scratch.

Where the efficiency comes from

The learning signal is richer: a full distribution over next tokens (KL divergence) instead of a single correct token (cross-entropy).

THE AGENDA · DISTILLATION-TIME INTERVENTIONS

Labs distill in two contexts; we focus on pre-training

IN SCOPE

Pre-training distillation

Take a freshly initialized small model. Instead of cross-entropy on the corpus, minimize KL divergence to the logits of a much larger teacher run over that corpus.

More compute-efficient than training the small model from scratch.

OUT OF SCOPE

Post-training distillation

For example, on-policy self-distillation: consolidating RL or post-RL checkpoints back into a common mid-trained model.

WHY DISTILLATION · 1 OF 2

Pre-training interventions are interventions on the learning signal

Data filtering

Zero out the cross-entropy loss on a subset of tokens in the corpus.

Gradient routing

Modular pre-training: control where in the model a gradient is allowed to cause updates.

Distillation

Swap cross-entropy for KL divergence on teacher logits.

KL divergence is a loss just like cross-entropy, so modifying it belongs to the **same family of signal-time interventions**, and the same levers should apply.

WHY DISTILLATION · 2 OF 2

It is also the practical place to intervene

01

Labs already do it, a lot

Small production models are routinely distilled from larger ones.

02

It is cheap to experiment with

Within reach of non-lab players like us, not only frontier labs.

03

It is a second chance

Apply the pre-training interventions that did not make it into the original teacher.

THEORY OF CHANGE

Make pre-training interventions cheap enough to actually get used

WHY IT MATTERS

Pre-training interventions control capabilities more deeply and robustly than post-training alone.

→

THE PROBLEM

They are neglected, and a big reason is that they are expensive.

→

OUR LEVER

Distillation-time versions are cheap to develop and cheap to apply.

→

THE CHANGE

More avenues and opportunities for developers to implement them.

RESEARCH BETS

The questions, in the order we'd tackle them

Core: methods

1. Can we apply data filtering while distilling A → B?

2. Does the naive version work: filter the data, otherwise distill as normal? If not, why not?

3. Can we modify the distillation objective so it works while keeping distillation's efficiency gain? What is the alignment tax?

4. Can gradient-routing techniques modularize a capability inside B during distillation?

Extensions

Is there an analog for latent distillation (MSE on activations instead of KL on logits)?

How does this connect to subliminal learning?

Does it work for subtle, overlapping concepts, or only for disjoint token sets like Japanese vs. English?

How robust is the result to adversarial fine-tuning?

THE LOAD-BEARING ASSUMPTION

Naive data filtering won't survive distillation. The teacher's logits on *retain* data are already contaminated with forget-set knowledge.

Why it is load-bearing

If it holds, filtering the distillation corpus is not enough, and the team must spend its first months on methods development.

What would make us pivot

Naive filtered distillation removes the forget capability with no significant contamination. Then skip methods work and go straight to scaling.

STAFFING

A team of four, research-heavy first

IN PLACE

Lead researcher (me)

50% hands-on research, 50% management. Sets direction across both tracks.

HIRE FIRST

Research IC · methods

Develops the modified distillation objective and its ablations.

HIRE FIRST

Research IC · scaling

Quantifies efficiency gains vs. cross-entropy and proves the method out at larger scales.

HIRE LATER

Research engineer

Most useful once we set up large scaling experiments.

MILESTONES

Close out methods by month 3, then scale

3 months

- Methods development closed out: a distillation objective that filters without losing the efficiency gain
- Efficiency vs. cross-entropy training quantified
- Both ICs move onto scaling research

12 months

- A paper ready for ICML or NeurIPS
- Scaling results in a toy setting and in realistic settings, ideally up to multiple billions of parameters
- Extensive ablations

RISKS

Some things are much harder to forget than others

EASIER

An entire language

Largely disjoint tokens

HARDER

A domain, like virology

Overlaps with general biology

HARDEST

A concept, like deception

Abstract, spread across all text

The risk: methods that work on part of this spectrum **don't generalize** to the more ambitious, abstract end.

PART 2 · THE DERISKING EXPERIMENT

Does forget-set knowledge leak through the teacher's logits, and can a cheap fix stop it?

A direct test of the load-bearing assumption, at toy scale.

EXPERIMENT · SETUP

A toy proxy: SimpleStories, one topic to forget

Dataset	SimpleStories: synthetic children's stories, labeled with 48 topics
Forget	1 of the 48 topics
Retain	The other 47 topics (also called "core")
Models	8 models: two at 20M parameters, six at 5M
Budget	Chinchilla-optimal: ~20 tokens per parameter

Why SimpleStories

Every document already has a topic label, so the filter is exact.

Small models trained on it are coherent.

I used it for methods development in my ICML paper, and those findings held at much larger scale.

EXPERIMENT · THE EIGHT MODELS

Four trained on tokens, four distilled on logits

Model	Size	Loss	Training data	Teacher
Base-20M	20M	Cross-entropy	All 48 topics	—
Filter-20M	20M	Cross-entropy	Retain (47)	—
Base-5M	5M	Cross-entropy	All 48 topics	—
Filter-5M	5M	Cross-entropy	Retain (47)	—
Distill-All-5M	5M	KL on logits	All 48 topics	Base-20M
Distill-Base-5M	5M	KL on logits	Retain (47)	Base-20M (naive filtering)
Distill-Filter-5M	5M	KL on logits	Retain (47)	Filter-20M (gold standard)
Distill-Diff-5M	5M	KL on logits	Retain (47)	**Base-20M, corrected (ours)**

EXPERIMENT · THE METHOD

Distill-Diff: subtract the forget direction from the teacher

$$z_{\text{target}} = z_{\text{base}} - \alpha \cdot \text{ReLU}(z_{\text{spec}} - z_{\text{base}})$$

$$\mathcal{L} = \tau^2 \cdot \text{KL}(\text{softmax}(z_{\text{target}} / \tau) \,\|\, \text{softmax}(z_{\text{student}} / \tau))$$

forward KL, on retain data only

z_{base} · the teacher

Logits of Base-20M, the unfiltered teacher we actually have.

z_{spec} · the specialist

Base-20M briefly fine-tuned on the forget topic. Cheap to produce.

α · ReLU(excess)

ReLU keeps only logits the specialist pushes above the teacher. α sets how hard we subtract them.

RESULTS · RAW LOSS

Raw cross-entropy is hard to read on its own



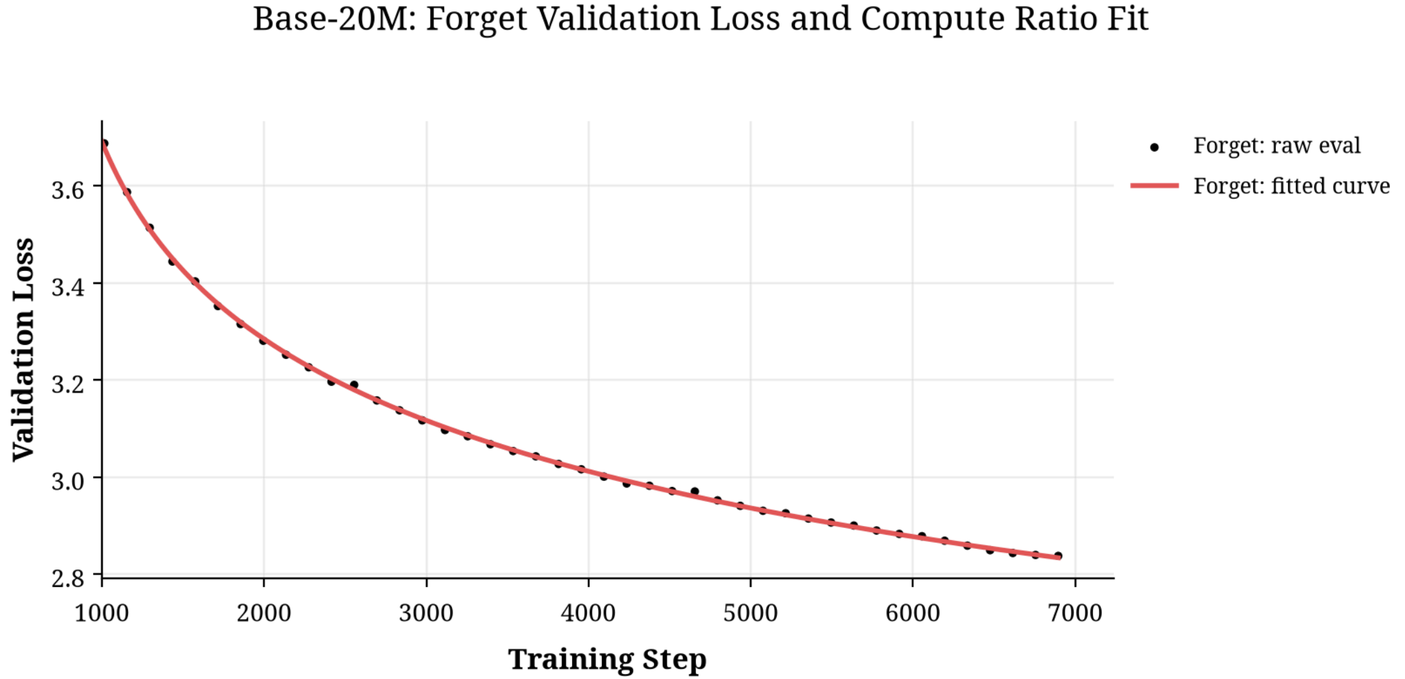
Validation loss on **retain** and **forget** data for all eight models.

The 5M models differ by a few hundredths of a nat. Is 3.806 vs. 3.826 a big gap or a small one?

We need a compute-adjusted scale.

RESULTS · THE METRIC

Compute ratio: loss as training compute



1. Record a baseline's validation loss through training.

2. Fit a power law in log space:
$L(s) = A \cdot (s + x_0)^{-\alpha}$

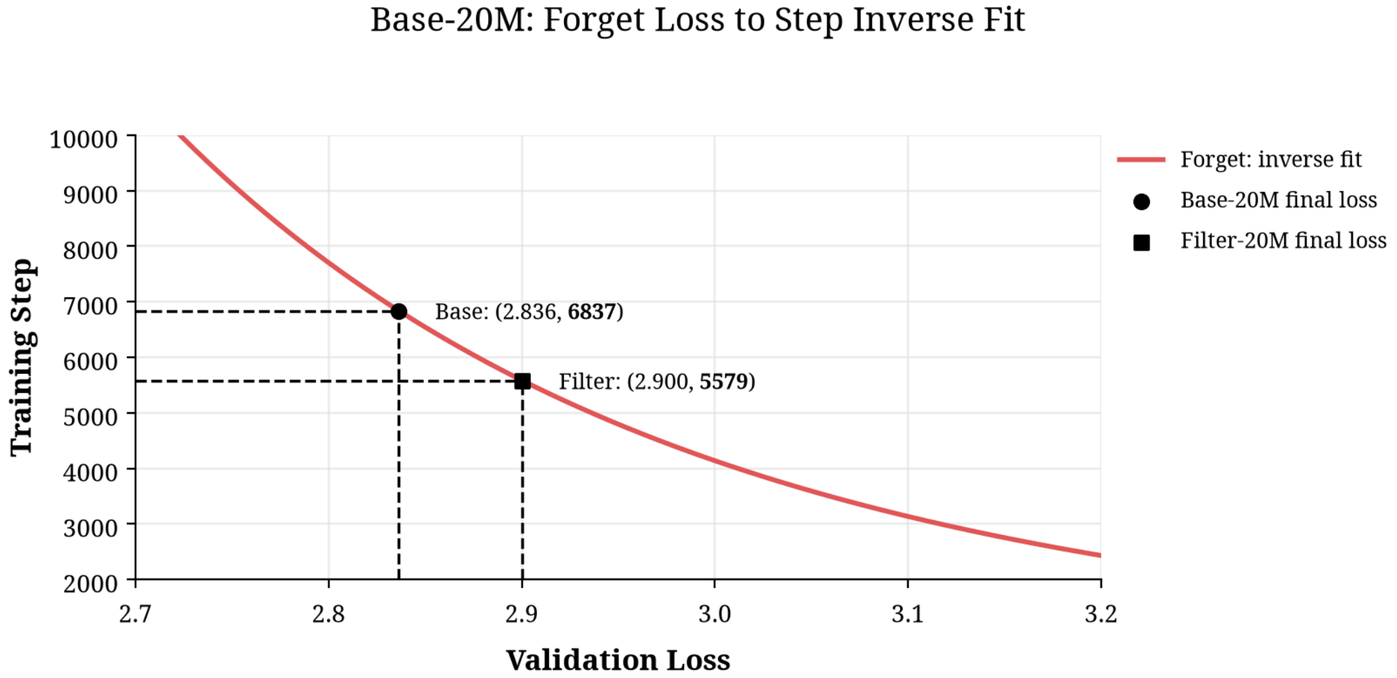
3. Invert it to map any loss to a baseline step.

4. Divide by the baseline's own step-equivalent.

0.9 = as good as the baseline was 90% of the way through training.

RESULTS · THE METRIC

Reading the inverse fit



Base-20M's final forget loss, 2.836, maps to step **6,837**.

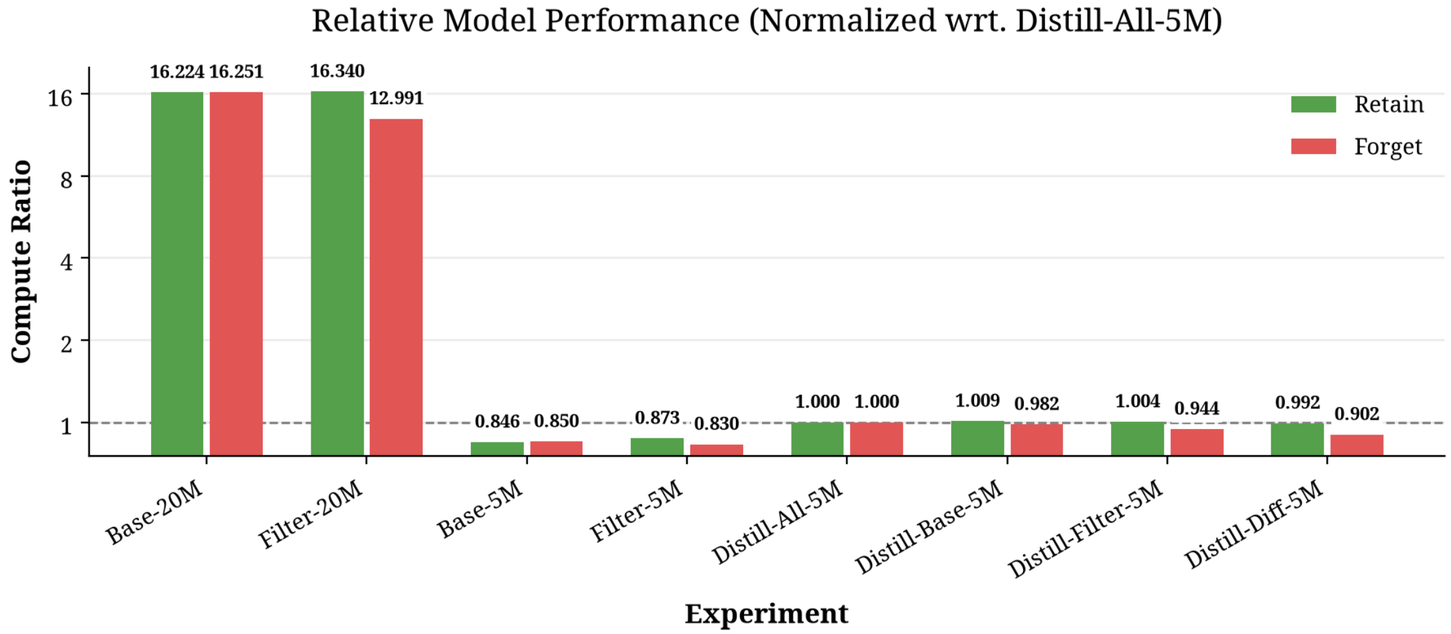
Filter-20M's forget loss, 2.900, is where Base-20M was at step **5,579**.

≈ 0.82

5,579 ÷ 6,837: filtering cost the teacher about a fifth of its forget-topic compute

RESULTS · SANITY CHECK

The metric recovers the 16× compute gap



All eight models, normalized to **Distill-All-5M** = 1.

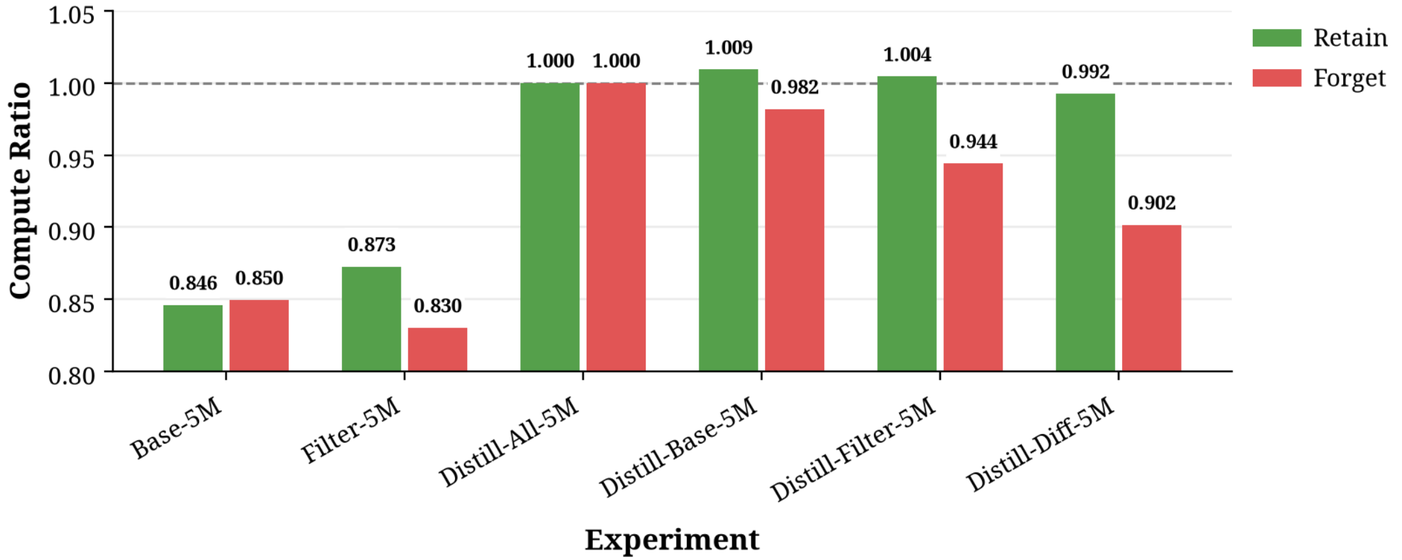
The 20M models land at **~16**. Chinchilla-optimal training gives them 4× the parameters × 4× the tokens = **16× the compute**.

We didn't tune for this, so the metric tracks real compute.

Values above 1 extrapolate the fitted power law past the baseline's training run.

RESULTS · THE 5M MODELS

Distill-Diff beats the gold standard on forgetting



Base / Filter-5M (cross-entropy)
~0.85: same data, less efficient than distilling

Distill-Base (naive filter)
Forget **0.982**: some effect, not strong

Distill-Filter (gold standard)
Forget **0.944**, but needs a filtered teacher

Distill-Diff (ours)
Retain **0.992**, forget **0.902**

WHAT WE FOUND

The assumption holds, and a cheap fix already works

0.85

Distillation is more efficient

Cross-entropy 5M models reach ~0.85 of Distill-All on the same data. The efficiency gain is real at this scale.

0.982

Naive filtering leaks

Distilling only on retain data barely suppresses the forget topic: the teacher's logits carry it.

0.902

Distill-Diff closes the gap

Forgets more than the filtered-teacher gold standard (0.944) at under 1% retain cost (0.992).

NEXT STEPS

From one toy result to a method that scales

Harden this result

Error bars across seeds before trusting few-percent gaps

Repeat with each of the 48 topics as the forget set

Measure the retain-side tax precisely, and tune how much of the difference to subtract

Test robustness to adversarial fine-tuning

Extend the agenda

Scale teacher and student sizes toward billions of parameters

Move up the difficulty spectrum: overlapping concepts, then WMDP-Bio

Try gradient routing and latent (MSE) distillation variants

APPENDIX

Code, data and substitutions

Code	[GitHub repo link]
Analysis	compile.py, load_data.py, plot.py: compute-ratio fit and charts
Data	SimpleStories [link]
Samples	[shared folder link]

What we'd normally use	Toy substitute
WMDP-Bio forget corpus	1 SimpleStories topic
>100B teacher, >1T tokens	20M teacher, Chinchilla-optimal
Billion-scale students	5M students
Capability benchmarks	Validation loss as compute ratio